Data Log

Set Up

- Imported 2019_Q2, 2019_Q3, 2019_Q4, and 2020_Q1 into schema cyclistic_case_study
- Created backup tables in schema cyclistic_case_study_backups
- Normalized column names. Table at end of log.
- Normalized usertype values based on report vocabulary. Member/Casual

```
--find usertype names
select usertype
from cyclistic_case_study."2019_Q3"
group by usertype;
--change usertype term for Member
update cyclistic_case_study."2020_Q1"
set usertype = replace(usertype, 'Subscriber', 'Member');
--change usertype term for Casual
update cyclistic_case_study."2020_Q1"
set usertype = replace(usertype, 'customer', 'Casual');
```

- Added trip_duration to "2020_Q1"

```
alter table cyclistic_case_study."2020_Q1"
add column trip_duration int4
```

- Computed and checked trip_duration for "2020_Q1"

```
--subtract start_time from end time to get seconds of trip
update cyclistic_case_study."2020_Q1"
set trip_duration = extract(epoch from end_time - start_time);
--show 5 values to check work
select end_time, start_time, trip_duration
from cyclistic_case_study."2020_Q1"
limit 5;
```

- Added null columns for gender and birthyear to "2020_Q1"

```
alter table cyclistic_case_study."2020_Q1"
add column birthyear int4,
add column gender varchar(50)
```

- Checked that all columns existed and data type was correct.

```
--Claude Sonnet 4.6 Assisted
select table_name, column_name, data_type
from information_schema.columns
where table_schema = 'cyclistic_case_study'
and table_name in ('2019_Q2', '2019_Q3', '2019_Q4', '2020_Q1')
and column_name in ('trip_id', 'start_time', 'end_time', 'start_station_name',
'start_station_id', 'end_station_name', 'end_station_id','usertype',
'trip_duration', 'gender', 'birthyear')
order by column_name, table_name;
```

- Aggregated trip_id, start_time, end_time, start_station_name_start_station_id, end_station_name, end_station_id, usertype, trip_duration, gender, and birthyear for all 4 quarters into single table, four_quarters

```
--Claude Sonnet 4.6 Assisted
create table cyclistic_case_study.four_quarters as

select trip_id, start_time, end_time, start_station_name, start_station_id,
end_station_name, end_station_id, usertype, trip_duration, gender, birthyear
from cyclistic_case_study."2019_Q2"
union all

select trip_id, start_time, end_time, start_station_name, start_station_id,
end_station_name, end_station_id, usertype, trip_duration, gender, birthyear
from cyclistic_case_study."2019_Q3"
union all

select trip_id, start_time, end_time, start_station_name, start_station_id,
end_station_name, end_station_id, usertype, trip_duration, gender, birthyear
from cyclistic_case_study."2019_Q4"
union all

select trip_id, start_time, end_time, start_station_name, start_station_id,
end_station_name, end_station_id, usertype, trip_duration, gender, birthyear
from cyclistic_case_study."2020_Q1";
```

- Added day_of_week column

```
--select extract(dow from start_time) as day,
alter table cyclistic_case_study.four_quarters
add column day_of_week varchar(50);

--Claude check before executing. "FMDay removes trailing spaces added by
Postgres."
update cyclistic_case_study.four_quarters
set day_of_week = to_char(start_time, 'FMDay');

--Result Check. look right? yes. okay good.
select *
from cyclistic_case_study.four_quarters fq
limit 20;
```

- Added age_demo column

```
alter table cyclistic_case_study.four_quarters
add column age_demo varchar(50);

--Claude Sonnet 4.6 Assisted. Find age and put into category, null for below 17 &
below 0.
update cyclistic_case_study.four_quarters
set age_demo = case
    when (2019 - birthyear) between 18 and 24 then '18-24'
    when (2019 - birthyear) between 25 and 34 then '25-34'
    when (2019 - birthyear) between 35 and 44 then '35-44'
    when (2019 - birthyear) between 45 and 54 then '45-54'
    when (2019 - birthyear) between 55 and 64 then '55-64'
    when (2019 - birthyear) >= 65 then '65+'
```

```
    else null
end;

--Result Check. look right? yes. okay good.
select birthyear, age_demo, (2019-birthyear) as calculated_age
from cyclistic_case_study.four_quarters fq
where birthyear is not null
limit 20
```

- Asked Claude what else might be useful to calculate. Recommendations: ~~month~~, ~~season~~, ~~hour_of_day~~, time_of_day, ~~is_weekend~~, ~~is_round_trip~~, trip_duration_bucket, ~~start_station_frequency~~
- Created time_of_day column. *Will consider trip_duration_bucket after init analysis of trip_duration*

```
--Claude Sonnet 4.6 Assisted. Might reconsider time categories to add early morning.
alter table cyclistic_case_study.four_quarters
add column time_of_day VARCHAR(20);

update cyclistic_case_study.four_quarters
set time_of_day = case
    when extract(hour from start_time) >= 5  and extract(hour from start_time) <12 then 'Morning'
    when extract(hour from start_time) >= 12 and extract(hour from start_time) < 17 then 'Afternoon'
    when extract(hour from start_time) >= 17 and extract(hour from start_time) < 21 then 'Evening'
    else 'Night'
end;
```

Data Validation

Checked number of casual vs member riders from four_quarters and unformatted datasets

```
--Claude Sonnet 4.6 Assisted. This melted my brain. One day. One day.
SELECT 'backup' AS source,
    CASE
        WHEN usertype IN ('Subscriber', 'member') THEN 'Member'
        WHEN usertype IN ('Customer', 'customer', 'casual') THEN 'Casual'
    END AS usertype,
    COUNT(*) AS count
FROM (
    SELECT "User Type" AS usertype FROM cyclistic_case_study_backups."2019_Q2 (Backup)"
    UNION ALL
    SELECT usertype FROM cyclistic_case_study_backups."2019_Q3 (Backup)"
    UNION ALL
    SELECT usertype FROM cyclistic_case_study_backups."2019_Q4 (Backup)"
    UNION ALL
    SELECT member_casual AS usertype FROM cyclistic_case_study_backups."2020_Q1 (Backup)"
) AS backup_data
GROUP BY
    CASE
        WHEN usertype IN ('Subscriber', 'member') THEN 'Member'
        WHEN usertype IN ('Customer', 'customer', 'casual') THEN 'Casual'
```

```
    END

UNION ALL

SELECT 'four_quarters' AS source, usertype, COUNT(*) AS count
FROM cyclistic_case_study.four_quarters
GROUP BY usertype

ORDER BY usertype, source;
```

Output:

Source	Usertype	count
backup	Casual	1291454
four_quarters	Casual	1291454
backup	Member	3525032
four_quarters	Member	3525032

Checked if values are consistent. (No Male from one set and male from another.)

```
select gender, count(*)
from cyclistic_case_study.four_quarters fq
group by gender
```

Output:

Gender	count
	760254
Female	1036071
Male	2593274
[NULL]	426887

[Null] is from 2020_Q1. “ ” is because gender is an option field. Need to change “ “ to NULL.

```
UPDATE cyclistic_case_study.four_quarters
SET gender = NULL
WHERE gender = '';
```

New Output:

Gender	count
Female	1036071
Male	2593274
[NULL]	1187141

Birthyear. Checking for ‘ ‘ vs NULL

```
select birthyear, count(*) as count
from cyclistic_case_study.four_quarters fq
group by birthyear
order by count desc
```

Output:

Birthyear	count
[NULL]	1164028
1992	236288

Noticed likely inaccurate birthyear dates

```
select birthyear, count(*) as count
from cyclistic_case_study.four_quarters fq
```

```
where birthyear < 1919
group by birthyear
order by birthyear
```

Output:

birthyear	count
1759	1
1790	1
1888	40
1889	6
1890	2
1899	170
1900	270
1901	19
1904	286
1905	6
1909	11
1910	6
1911	1
1912	2
1918	19

840 riders older than 100! Medical technology has come a long way.

birthyear	count
2002	2871
2003	1105
2014	5

3,976 riders under 18, not counting the 5 5-year-olds.

Revise age bracket to change birthyears before 1919 to NULL and include 16+ to lowest age bracket.

```
update cyclistic_case_study.four_quarters
set birthyear = case
       when birthyear < (1919) then null
       else birthyear
end;

update cyclistic_case_study.four_quarters
set age_demo = case
    when (2019 - birthyear) between 16 and 24 then '16-24'
    when (2019 - birthyear) between 25 and 34 then '25-34'
    when (2019 - birthyear) between 35 and 44 then '35-44'
    when (2019 - birthyear) between 45 and 54 then '45-54'
    when (2019 - birthyear) between 55 and 64 then '55-64'
    when (2019 - birthyear) >= 65 then '65+'
    else null
end;
```

Checking age_demo

```
select age_demo, count(*)
from cyclistic_case_study.four_quarters fq
group by age_demo
```

age_demo	count
16-24	437627
25-34	1883888
35-44	708863
45-54	360474
55-64	218839
65+	41922
[NULL]	1164873

Critical Error:

*****I mistakenly uploaded Q3 data to both 2019_Q3 and 2019_Q4. I discovered this while working in Tableau. I should have double checked the import values after uploading the data. Analysis below is invalid but will remain for posterity. Correct values are given below. *****

Baseline Analysis

~~Full Report~~

```
select var, count(*), round(count(*) * 100.0 / sum(count(*)) over (), 2) as
"%"
from cyclistic_case_study.four_quarters
group by var
```

~~usertype~~	~~count~~	~~%~~
~~Casual~~	~~1291454~~	~~26.81~~
~~Member~~	~~3525032~~	~~73.19~~

~~gender~~	~~Count~~	~~%~~
~~Female~~	~~1036071~~	~~21.51~~
~~Male~~	~~2593274~~	~~53.84~~
~~[NULL]~~	~~1187141~~	~~24.65~~

~~age_demo~~	~~count~~	~~%~~
~~16-24~~	~~437627~~	~~9.09~~
~~25-34~~	~~1883888~~	~~39.11~~
~~35-44~~	~~708863~~	~~14.72~~
~~45-54~~	~~360474~~	~~7.48~~
~~55-64~~	~~218839~~	~~4.54~~
~~65+~~	~~41922~~	~~0.87~~
~~[NULL]~~	~~1164873~~	~~24.19~~

~~day_of_week~~	~~count~~	~~%~~
~~Friday~~	~~710845~~	~~14.76~~
~~Monday~~	~~703649~~	~~14.61~~
~~Saturday~~	~~663455~~	~~13.77~~
~~Sunday~~	~~557862~~	~~11.58~~

Thursday	727844	15.11
Tuesday	715586	14.86
Wednesday	737245	15.31

time_of_day	count	%
Afternoon	1571001	32.62
Evening	1455317	30.22
Morning	1428834	29.67
Night	361334	7.50

Usertype Contrast

```
--Claude assist to find partition by to separate % pools by usertype.
select usertype, var, count(*), round(count(*) * 100.0 / sum(count(*)) over
(partition by usertype), 2) as "%"
from cyclistic_case_study.four_quarters fq
group by usertype, var
```

usertype	gender	count	%
Casual	Female	198598	15.38
Casual	Male	309107	23.93
Casual	[NULL]	783749	60.69
Member	Female	837473	23.76
Member	Male	2284167	64.80
Member	[NULL]	403392	11.44

Usertype	age_demo	count	%
Casual	16-24	131487	10.18
Casual	25-34	253254	19.61
Casual	35-44	70648	5.47
Casual	45-54	38569	2.99
Casual	55-64	17469	1.35
Casual	65+	2052	0.16
Casual	[NULL]	777975	60.24
Member	16-24	306140	8.68
Member	25-34	1630634	46.26
Member	35-44	638215	18.11
Member	45-54	321905	9.13
Member	55-64	201370	5.71
Member	65+	39870	1.13
Member	[NULL]	386898	10.98

usertype	day_of_week	count	%
Casual	Friday	178285	13.80
Casual	Monday	150892	11.68
Casual	Saturday	305387	23.65
Casual	Sunday	241714	18.72
Casual	Thursday	152616	11.82
Casual	Tuesday	127470	9.87
Casual	Wednesday	135090	10.46

Member	Friday	532560	15.11
Member	Monday	552757	15.68
Member	Saturday	358068	10.16
Member	Sunday	316148	8.97
Member	Thursday	575228	16.32
Member	Tuesday	588116	16.68
Member	Wednesday	602155	17.08

Usertype	time_of_day	count	%
Casual	Afternoon	556060	47.06
Casual	Evening	359807	27.86
Casual	Morning	252815	19.58
Casual	Night	122772	9.51
Member	Afternoon	1014941	28.79
Member	Evening	1095510	31.08
Member	Morning	1176019	33.36
Member	Night	238562	6.77

Trip Duration:

```
select round(avg(trip_duration)/60, 2) as average_trip_duration
from cyclistic_case_study.four_quarters
```

Usertype	average_trip_duration
All	26.83
Casual	59.33
Member	14.92

Q4 Data Fix.

- Uploaded correct Q4 data this time as table, 2019_Q4_correct.
- Dropped incorrect tables and replaced them as 2019_Q4 and 2019_Q4 (Backup)

```
drop table cyclistic_case_study_backups."2019_Q4 (Backup)";

alter table cyclistic_case_study_backups."2019_Q4_correct"
rename to "2019_Q4 (Backup)";

drop table cyclistic_case_study."2019_Q4";

create table cyclistic_case_study."2019_Q4" as
select * from cyclistic_case_study_backups."2019_Q4 (Backup)";
```

- Normalized 2019_Q4 Data. Renamed Columns.

```
--change usertype term for Member
update cyclistic_case_study."2019_Q4"
set usertype = replace(usertype, 'Subscriber', 'Member');
--change usertype term for Casual
update cyclistic_case_study."2019_Q4"
set usertype = replace(usertype, 'Customer', 'Casual');
```

- Renamed four_quarters to bad_four_quarters. Started rebuilding four quarters table with same queries from above. This whole fix took around 20 minutes since everything was very organized. Thank you past me for at least getting that right.

Data Validation Part 2: Electric Boogaloo

Source	usertype	count
backup	Casual	905954
four_quarters	Casual	905954
backup	Member	2973868
four_quarters	Member	2973868

gender	count
Female	791060
Male	2122380
[NULL]	966382

age_demo	count
16-24	337661
25-34	1487946
35-44	585344
45-54	300326
55-64	184287
65+	36638
[NULL]	947620

Baseline Analysis

Full Report

usertype	count	%
Casual	905954	23.35
Member	2973868	76.65

gender	count	%
Female	791060	20.39
Male	2122380	54.70
[NULL]	966382	24.91

age_demo	count	%
16-24	337661	8.70
25-34	1487946	38.35
35-44	585344	15.09
45-54	300326	7.74
55-64	184287	4.75
65+	36638	0.94
[NULL]	947620	24.42

day_of_week	count	%
Friday	575723	14.84
Monday	576648	14.86
Saturday	497501	12.82
Sunday	449271	11.58
Thursday	587524	15.14
Tuesday	599636	15.46
Wednesday	593519	15.30

time_of_day	count	%
Afternoon	1266863	32.65
Evening	1141114	29.41
Morning	1200949	30.95
Night	270896	6.98

Usertype Contrast

usertype	gender	count	%
Casual	Female	129564	14.30
Casual	Male	208683	23.03
Casual	[NULL]	567707	62.66
Member	Female	661496	22.24
Member	Male	1913697	64.35
Member	[NULL]	398675	13.41

usertype	age_demo	count	%
Casual	16-24	87210	9.63
Casual	25-34	169598	18.72
Casual	35-44	47626	5.26
Casual	45-54	25020	2.76
Casual	55-64	11337	1.25
Casual	65+	1389	0.15
Casual	[NULL]	563774	62.23
Member	16-24	250451	8.42
Member	25-34	1318348	44.33
Member	35-44	537718	18.08
Member	45-54	275306	9.26
Member	55-64	172950	5.82
Member	65+	35249	1.19
Member	[NULL]	383846	12.91

usertype	day_of_week	count	%
Casual	Friday	122933	13.57
Casual	Monday	104452	11.53
Casual	Saturday	209543	23.13
Casual	Sunday	181299	20.01
Casual	Thursday	103347	11.41

Casual	Tuesday	91191	10.07
Casual	Wednesday	93189	10.29
Member	Friday	452790	15.23
Member	Monday	472196	15.88
Member	Saturday	287958	9.68
Member	Sunday	267972	9.01
Member	Thursday	484177	16.28
Member	Tuesday	508445	17.10
Member	Wednesday	500330	16.82

usertype	time_of_day	count	%
Casual	Afternoon	399181	44.06
Casual	Evening	242219	26.74
Casual	Morning	182958	20.20
Casual	Night	81596	9.01
Member	Afternoon	867682	29.18
Member	Evening	898895	30.23
Member	Morning	1017991	34.23
Member	Night	189300	6.37

Trip Duration:

usertype	average_trip_duration
All	24.62
Casual	14.16
Member	58.96

2019_Q2	2019_Q3	2019_Q4	2020_Q1	Standarized Name
01 - Rental Details Rental ID *int4*	trip_id *int4*	trip_id *int4*	ride_id (varchar(50))	trip_id (varchar(50))
			rideable_type	
01 - Rental Details Local Start Time	start_time	start_time	started_at	start_time
01 - Rental Details Local End Time	end_time	end_time	ended_at	end_time
03 - Rental Start Station Name	from_station_name	from_station_name	start_station_name	start_station_name
03 - Rental Start Station ID	from_station_id	from_station_id	start_station_id	start_station_id
03 - Rental End Station Name	to_station_name	to_station_name	end_station_name	end_station_name
03 - Rental End Station ID	to_station_id	to_station_id	end_station_id	end_station_id
			start_lat	
			start_lng	
			end_lat	
			end_lng	
User Type	usertype	usertype	member_casual	usertype
01 - Rental Details Duration In Seconds Uncapped	tripduration	tripduration	ride_duration	trip_duration
Member Gender	gender	gender	Null	gender
05 - Member Details Member Birthday Year	birthyear	birthyear	null	birthyear
02 - Rental Details Bike ID	bikeid	bikeid		bikeid